Table 1: Literature related to metrology sampling and monitoring decisions.

Reference	Title	Topic	Relevance to Continuous Probing
Nduhura-Munga (2013)	A Literature Review on Sampling Techniques in Semiconductor Manufacturing	Sampling strategies	Survey of static, adaptive, and dynamic sampling strategies. Sampling frequency and measurement coverage naturally appear as controllable measurement intensities.
Susto et al. (2019)	Induced Start Dynamic Sampling for Wafer Metrology	Dynamic sampling	Adaptive strategy selecting wafer measurement sites to reduce metrology cost while maintaining reconstruction accuracy.
Susto et al. (2018)	A Dynamic Sampling Approach for Cost Reduction in Semiconductor Manufacturing	Dynamic metrology sampling	Optimizes measurement effort across wafers to balance monitoring accuracy and metrology capacity.
McLoone et al. (2018)	Efficient Dynamic Spatial Sampling and Reconstruction of Wafer Profiles	Spatial sampling	Reduces the number of spatial measurements required for wafer profile reconstruction, controlling measurement coverage.
Orji et al. (2018)	Metrology for the Next Generation of Semiconductor Devices	Metrology review	Discusses trade-offs between measurement precision, throughput, and cost in semiconductor metrology systems.
Kang et al. (2011)	Virtual Metrology for Run-to-Run Control in Semiconductor Manufacturing	Virtual metrology	Uses predictive models to estimate process outputs when measurements are unavailable or costly.
Maitra (2024)	Review: Virtual Metrology in Semiconductor Manufacturing	Virtual metrology survey	Reviews predictive monitoring approaches that reduce dependence on frequent physical measurements.
Bode (2002)	Run-to-Run Control and Performance in Semiconductor Manufacturing	Run-to-run control	Analyzes performance of process control strategies when measurements are intermittent.
Musacchio (1997)	On the Utility of Run-to-Run Control in Semiconductor Manufacturing	Process control theory	Early analysis of control policies under limited measurement information.
Prabhu (2008)	Performance Monitoring of Run-to-Run Control Systems	Process monitoring	Examines monitoring strategies when measurement resources are limited.
Dreyfus (2022)	Virtual Metrology as an Approach for Product Quality Monitoring	Predictive monitoring	Demonstrates how predictive models substitute for frequent metrology measurements.
Bao (2013)	Run-to-Run Controller for Product Surface Quality	Adaptive control	Control performance depends on the availability and precision of process measurements.
Jebri / Jeon (2017)	Virtual Metrology Applied in Run-to-Run Control	Virtual metrology integration	Integrates predictive metrology models with run-to-run process control.
Chen (2025)	Review of Static, Adaptive, and Dynamic Sampling in Semiconductor Manufacturing	Sampling strategies	Discusses adaptive and dynamic sampling methods that adjust measurement frequency based on process conditions.

Reference	Title	Topic	Relevance to Continuous Probing
Statistical Methods for Semiconductor Manufacturing	Statistical Methods for Semiconductor Manufacturing	Statistical process monitoring	Describes statistical monitoring techniques when measurement data are limited.
Semiconductor Industry Association	Metrology in Semiconductor Manufacturing	Industry overview	Highlights throughput–precision tradeoffs that motivate flexible measurement strategies.

1 Simulation Model and Experimental Setup

We consider a two-stage service system with a front-end (FE) and two alternative back-end (BE) resources: a fast server and a standard server.

Jobs arrive sequentially with stochastic interarrival times. Each job n has a demand level $d_n \in [0,1]$, which determines revenue, and belongs to a class $j \in \{1,\dots,m\}$. Conditional on its class, the job is complex with a class-dependent probability. Complexity affects processing at the standard BE through potential disruptions.

Each job is processed through FE, followed by a transfer delay, and then routed to either the fast or the standard BE. The fast BE is typically faster but incurs an additional cost, while the standard BE may experience delays if complex jobs trigger extra processing.

Policies We consider a finite set of routing policies $\{x_1,\dots,x_k\}$. Each policy is a parametric decision rule of the form

$$\text{score} = a_d \cdot d_n - a_q \cdot (Q_n^{\text{fast}} - Q_n^{\text{std}}) + a_s \cdot s_n,$$

and the job is routed to the fast BE if score $\geq \tau$. Here, Q_n^{fast} and Q_n^{std} denote the queueing delays at the time of routing, and s_n is an information signal (if available).

Probing A probing decision $z \in [0,1]^m$ specifies the inspection effort applied to each class.

Probing affects system performance in two ways:

- It generates a noisy signal about job complexity, which can be used in routing decisions.
- It reduces the likelihood and/or severity of disruptions at the standard BE.

Signals are generated stochastically as a function of the job class, its true complexity, and the probing level. When no signal is observed, a class-dependent baseline value is used.

Performance Let $Y(x_i,\xi;z)$ denote the total profit under policy x_i, randomness ξ, and probing decision z. Profit accounts for revenue (increasing in demand), processing costs at FE and BE, transfer costs, and penalties for tardiness.

Two-stage evaluation For each probing decision z, performance is evaluated using a two-stage simulation procedure. First, the best policy is selected from the candidate set using a finite number of simulation replications (training stage). Then, the selected policy is evaluated using an independent set of replications (evaluation stage) to obtain an unbiased estimate.

Let $\hat{i}(z)$ denote the selected policy. The value of probing decision z is

$$Q(z) = \mathbb{E}\big[Y(x_{\hat{i}(z)},\xi;z)\big].$$

Objective The probing problem is

$$\max_{z\in[0,1]^m} \left\{-\alpha^\top z + Q(z)\right\}.$$

1.1 Model Variants

We consider a sequence of simulation models of increasing complexity.

SimOpt001: Baseline Jobs are identical apart from stochastic demand and a fixed probability of being complex. No class structure or information is available. Policies depend only on demand and queue differences. This model serves as a baseline for system behavior without heterogeneity or information.

SimOpt002: Heterogeneous classes without information Jobs belong to different classes, which determine their probability of being complex. However, class membership is not used in decision making, and no additional information is available. This introduces hidden heterogeneity without improving decisions.

SimOpt003: Perfect information benchmark The complexity of each job is perfectly observed and used directly in the routing decision. This provides an upper bound on performance and quantifies the value of full information.

SimOpt004: Probing model Probing determines the amount of information available about job complexity. For each job, a signal is generated probabilistically as a function of its class, its true complexity, and the probing level. The signal is noisy and may be absent. When absent, a baseline estimate is used.

In addition to generating information, probing also reduces the probability or impact of disruptions when complex jobs are processed at the standard BE. Thus, probing affects both decision quality and system dynamics.

SimOpt005: Two-stage evaluation The final model incorporates a two-stage evaluation procedure to correctly estimate the value of probing. For each probing decision, policies are first ranked and selected using a training set of simulation replications, and the selected policy is then evaluated on an independent test set. This structure avoids selection bias and yields reliable comparisons across probing decisions.

2 Continuous Probing via Inspection and Measurement Quality

We model probing as the allocation of inspection and metrology effort across job classes. The decision variable is

$$z \in [0,1]^m,$$

where z_j represents the inspection intensity applied to jobs of class j.

Higher values of z_j correspond to: (i) more frequent inspection (higher sampling rate), and/or (ii) higher-quality measurements (lower noise or more reliable classification).

2.1 Operational interpretation

In semiconductor manufacturing, inspection and metrology are used to generate signals about the downstream processing requirements or risk of a lot. These activities are costly and capacity-consuming, so they are applied selectively.

We interpret z_j as controlling both:

- the probability that a class-j job is inspected,
- the quality of the signal produced by inspection.

2.2 Signal model

Each job n belongs to a class $\eta_n \in \{1, \ldots, m\}$. When a job arrives at the decision point (e.g., before back-end routing), the system may generate an inspection signal.

For class j, the signal generation is governed by z_j as follows.

Inspection frequency. With probability z_j, the job is inspected and a signal is observed; with probability $1 - z_j$, no signal is generated:

$$S_n = \begin{cases} \tilde{S}_n, & \text{with probability } z_j, \\ \varnothing, & \text{with probability } 1 - z_j. \end{cases}$$

Signal quality. Conditional on inspection, the observed signal $\tilde{S}_n$ is a noisy indicator of the job's downstream processing characteristics.

We model $\tilde{S}_n$ as a continuous score:

$$\tilde{S}_n = \mu_j + \varepsilon_n,$$

where μ_j is a class-dependent baseline and

$$\varepsilon_n \mid \eta_n = j, z_j \sim \mathcal{N}(0, \sigma_j^2(z_j)).$$

The variance $\sigma_j^2(z_j)$ is decreasing in z_j, reflecting that higher inspection effort yields more precise and reliable measurements.

A typical specification is

$$\sigma_j^2(z_j) = \overline{\sigma}_j^2 e^{-\beta_j z_j},$$

where $\overline{\sigma}_j^2$ is the variance with minimal inspection and β_j controls the effectiveness of inspection effort.

Remarks. This formulation avoids introducing latent variables: the signal $\tilde{S}_n$ is directly generated from the class-dependent distribution, and probing affects only the frequency and precision of this observable signal.

2.3 Policy

Routing or dispatching decisions use the observed signal when available. For example,

$$\text{score}_n = a_d d_n - a_q\big(Q_n^{\text{fast}} - Q_n^{\text{std}}\big) + a_s\, \hat{S}_n,$$

where

$$\hat{S}_n = \begin{cases} \tilde{S}_n, & \text{if inspected,} \\ \bar{\mu}_j, & \text{if not inspected,} \end{cases}$$

and $\bar{\mu}_j$ is a default class-based estimate.

The job is routed to a fast or protected back-end resource if $\text{score}_n \geq \tau$.

2.4 System performance

Let $Y(x_i, \xi(z))$ denote the performance of the system under policy x_i when inspection decisions are given by z. The randomness $\xi(z)$ includes both the underlying system uncertainty and the stochastic signal generation process induced by inspection.

2.5 Optimization problem

The continuous probing problem is

$$\max_{z \in [0,1]^m} \left\{ -\alpha^\top z + \max_{i=1,\ldots,k} \mathbb{E}\big[Y(x_i, \xi(z))\big] \right\},$$

where α_j represents the cost of inspection effort for class j.

2.6 Interpretation

In this formulation:

- $z_j = 0$ corresponds to no inspection of class j;
- increasing z_j raises the probability of inspection and improves signal precision;
- inspection is costly through $\alpha^\top z$;
- the value of inspection arises from better routing and dispatching decisions enabled by more frequent and more reliable signals.

2.7 Discussion

This model captures a key operational trade-off: inspection and metrology improve decision quality but consume resources. The decision maker must allocate inspection effort across job classes to balance these costs against the benefits of improved system performance.

2.8 A simpler alternative

A simpler formulation models inspection as directly generating a noisy classification signal whose accuracy depends on the inspection intensity.

For a job of class $\eta_n = j$, the system produces a signal

$$S_n \in \{0, 1\},$$

where $S_n = 1$ may be interpreted as a warning or high-risk flag.

The distribution of S_n depends on z_j:

$$\mathbb{P}(S_n = 1 \mid \eta_n = j) = p_j(z_j),$$

where $p_j(z_j)$ varies with z_j, reflecting that inspection changes the reliability and bias of the signal.

The function $p_j(z_j)$ captures how inspection effort affects the distribution of the observed signal. In the absence of inspection, $p_j(0) = \bar{p}_j$ reflects a baseline flagging rate based on limited information. As z_j increases, inspection improves the reliability of the signal, which we model through a monotone function $p_j(z_j)$.

A typical specification is

$$p_j(z_j) = \bar{p}_j + (p_j^{\max} - \bar{p}_j)\left(1 - e^{-\beta_j z_j}\right),$$

which captures diminishing returns in inspection effort. Depending on the class, inspection may increase or decrease the probability of flagging, reflecting improved detection of problematic jobs or reduced false alarms for well-behaved jobs.

More generally, one may define a class-dependent signal distribution

$$S_n \sim \mathcal{P}_j(\cdot; z_j),$$

such that the signal becomes more informative as z_j increases (e.g., lower misclassification rates or clearer separation across classes).

Policy. The routing or dispatching policy uses the observed signal directly. For example,

$$\text{score}_n = a_d d_n - a_q\left(Q_n^{\text{fast}} - Q_n^{\text{std}}\right) + a_s S_n,$$

or more generally, policies may depend on the realized signal value without requiring explicit probabilistic inference.

Interpretation. In this formulation, inspection improves decision making by producing more reliable warning signals. Higher values of z_j increase the accuracy of these signals, enabling better routing and dispatching decisions without introducing latent variables or posterior beliefs.